

Hybrid Projection Methods for Multilingual Zero-shot Text Classification

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Abstract. Automatic text classification for languages with scarce labeled data poses significant challenges for real-world applications. Zero-shot learning circumvents this by transferring knowledge from high-resource to low-resource languages without requiring labeled target data. This paper presents a hybrid projection framework that aligns multilingual text embeddings with label embeddings to improve zero-shot performance on English, Hindi, and Bengali news articles. Our method uses LLaMA-3.1 and BLOOM-7B1 in a prompt-based setup alongside XLM-RoBERTa and IndicBERT for contextual embedding generation. These text embeddings are mapped into the semantic space of ConceptNet, GloVe, or GoogleNews label embeddings through ridge regression, Kolmogorov Arnold Networks, and support vector regression, optionally enriched by a fewshot augmentation. Experiments on three monolingual news datasets demonstrate that our approach achieves 79.87% accuracy for English, 60.50% for Hindi, and 41.71% for Bengali using a KAN-SVR hybrid. Compared to large language model baselines and entailment-based methods, our framework reduces the performance gap between high- and low-resource languages by over fifty percent, setting a new state-of-the-art in multilingual zero-shot text classification while maintaining computational efficiency and modular extensibility.

Keywords: Zero-Shot Classification, Large Language Models, Natural Language Processing, Artificial Intelligence

1 Introduction

The rapid expansion of digital content across multiple languages has made automated text classification indispensable for organizing and retrieving information [8]. While high-resource languages benefit from large annotated corpora, low-resource languages such as Hindi and Bengali suffer from scarce labeled data and underrepresentation in pretraining [15]. Additionally, rich morphological

complexity and inconsistent script encodings in these languages pose further obstacles to effective tokenization and embedding.

Zero-shot learning enables models to classify instances from unseen categories by leveraging semantic relationships between seen and unseen classes [34]. Prior work has demonstrated the efficacy of knowledge graph embeddings for social media classification [4], and recent benchmarks have evaluated entailment-based zero-shot approaches [35]. However, applying these techniques to multilingual news classification, particularly for low-resource Indian languages, remains underexplored. To address these gaps, we propose a hybrid projection framework that aligns multilingual text embeddings with label embeddings to improve zero-shot classification performance across English, Hindi, and Bengali news articles. Our contributions are as follows:

1. We introduce the first comprehensive study of zero-shot text classification for Hindi and Bengali news articles along with improvement on English news articles.
2. We develop novel hybrid projection methods combining KolmogorovArnold Networks (KAN) and Support Vector Regression (SVR) to capture complex semantic mappings.
3. We conduct extensive experiments comparing multiple embedding strategies and knowledge sources, including ConceptNet [29] and GloVe [24].
4. We propose a few-shot augmentation technique that further enhances performance by incorporating a small number of labeled examples.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the challenges of low-resource languages. Section 4 describes the datasets and proposed methodology. Section 5 reports experimental results and analysis. Section 6 concludes and outlines future work.

2 Related Work

The evolution of zero-shot text classification began with the fundamental limitations of traditional supervised methods [22], which required extensive labeled data for each target category. This constraint motivated the development of zero-shot learning approaches that could classify text into previously unseen categories [34], with early works [31] demonstrating promising results. The emergence of large pre-trained language models further enhanced these capabilities, with recent studies showing impressive performance in English text classification. Extending these successes to multilingual scenarios revealed significant challenges. While cross-lingual models [33] such as mBERT [8] and XLM-R [5] showed promise for high-resource languages, their performance degraded substantially for low-resource languages like Hindi and Bengali [16]. Studies by [15] and [14] highlighted specific challenges in Indian language processing, revealing that simple cross-lingual transfer was insufficient.

To address semantic gaps, researchers turned to structured knowledge. Knowledge graphs emerged as powerful resources [4], with ConceptNet [29] proving particularly influential for multilingual applications. Studies by [38] demonstrated

how knowledge graph embeddings could bridge language differences through structured relational information. This led to sophisticated integration methods that merged symbolic knowledge with neural representations, involving entity linking and graph convolutional techniques to enhance semantic representations and resolve multilingual ambiguities.

Parallel developments focused on improving text-label mapping mechanisms. Early projection methods included linear transformations and neural networks inspired by the Kolmogorov-Arnold representation theorem [17]. Recognizing single-method limitations, researchers developed ensemble approaches [37] that combined multiple projection techniques, optimizing diverse distance metrics to capture complex semantic relationships and improve classification robustness.

The limitations of pure zero-shot setting motivated exploration of few-shot learning [10], where minimal labeled examples significantly enhanced performance [1]. The combination of few-shot learning with knowledge graphs [3] offered new possibilities for low-resource languages. Emerging meta-learning strategies enabled rapid adaptation to new tasks with minimal supervision, incorporating prompt-based fine-tuning and self-supervised pre-training to narrow performance gaps across languages. These developments culminated in sophisticated transfer learning approaches [27,13] with various transfer strategies [6]. Studies by Kakwani et al. [15] and Khanuja et al. [16] specifically addressed Indian language challenges, contributing valuable insights about language-specific adaptations. This progression from simple zero-shot approaches to hybrid methods incorporating knowledge graphs, ensemble projections, and meta-learning strategies represents a natural evolution where each advancement addressed specific limitations while building toward more comprehensive multilingual solutions.

3 Challenges with Indian Languages

The advancement of zero-shot approaches to hybrid methods needs to address unique challenges in zero-shot text classification [16,15] in Indian languages such as Hindi and Bengali. These challenges can be categorized into several key areas.

Morphological Complexity Hindi and Bengali exhibited rich morphological structures with extensive inflections and derivations [14]. Unlike English, where word embeddings can effectively capture semantic relationships, these languages require sophisticated representation approaches due to their complex morphological structures. The highly inflectional nature of these languages results in a much larger vocabulary space, making it difficult for embedding models to capture all semantic variations effectively. Additionally, the agglutinative features of these languages, where multiple morphemes combine to form complex words, create challenges for tokenization and embedding generation.

Resource Scarcity Despite being widely spoken languages, Hindi and Bengali face significant resource constraints in terms of their pre-trained models and

knowledge bases. While ConceptNet includes entries for these languages, its coverage is substantially lower than that of English. Our research addresses this through the innovative use of multilingual embeddings (multilingual-MiniLM-L12) combined with English knowledge bases.

Script and Encoding Issues Hindi and Bengali scripts present additional challenges in text processing. Character encoding inconsistencies and the lack of standardization in digital texts can affect model performance [18]. Our pre-processing pipeline handles these script-specific challenges to ensure a consistent input representation.

3.1 Addressing the Challenges

Here, the primary task is to perform zero-shot text classification for news articles across English, Hindi, and Bengali languages. The goal is to categorize articles into their respective classes without using labeled examples from these classes during training. For English, we aim to classify articles into five categories: business, entertainment, politics, sports, and technology. For Bengali, the task involves six categories: Kolkata, state, national, sports, entertainment, and international news. For Hindi, we work with fourteen categories including india, international, entertainment, and sports news. This research addresses these challenges using several innovative approaches:

1. **Hybrid Projection Methods:** We introduce novel combinations of KAN and SVR with weighted predictions (90%+10% for Hindi), achieving significant improvements over baseline methods. The KAN component provides a powerful non-linear mapping that captures complex morphological relationships in Indian languages, while SVR offers stability and regularization. The weighted ensemble approach (optimized through cross-validation) combines their complementary strengths, leveraging expressive power of KAN while mitigating overfitting through regularization properties of SVR.
2. **Few-shot Augmentation:** By incorporating limited examples (10 instances per class), our approach partially compensates for the resource scarcity while maintaining the zero-shot learning framework’s advantages. These few examples serve as anchors in the embedding space, helping to establish clearer decision boundaries between classes. For morphologically rich languages, these examples help bridge the gap between abstract label representations and the diverse linguistic expressions found in actual news articles.
3. **Embedding Alignment:** Our methodology explores various combinations of embeddings and knowledge sources to find optimal representations for Indian languages, as demonstrated by our experimental results with different embedding combinations. Specifically, we align multilingual text embeddings with ConceptNet knowledge embeddings through projection learning, enabling better semantic matching between text and labels across languages. For Hindi and Bengali, this alignment process helps capture cultural and linguistic nuances that might be missed in standard multilingual embeddings.

The effectiveness of these approaches varies between Hindi and Bengali, with Hindi showing the most promising results. This variation suggests that language-specific adaptations may be necessary for optimal performance in various Indian languages.

4 Methodology

Various Zero-shot classification methods with certain parameters have been applied on complete news texts in different sub-continental and colonial languages. So, datasets carries utmost importance. Here various characteristics of dataset have been discussed.

4.1 Description of Datasets

The three monolingual news datasets used in this research are thematically related, allowing for meaningful cross-lingual performance comparison. All three datasets consist of news articles that cover broadly similar domains (such as politics, sports, entertainment), though with varying category granularity. The BBC News (English) [7] and BBC Hindi News datasets [23] are from the same source organization, ensuring consistent journalistic standards and topic coverage. The Bengali news dataset [28], while from a different source, maintains similar journalistic characteristics. This thematic alignment enables comparative evaluation of our zero-shot classification methods across languages despite the differences in category distribution and article length.

For our experimental evaluation, we normalized the categories across datasets where possible, matching conceptually similar classes (e.g., "sports" appears in all three datasets). The datasets also share temporal proximity, covering news content from similar time periods, which further supports meaningful cross-lingual comparison of classification performance.

This research utilizes three monolingual news datasets in the English, Bengali, and Hindi languages for zero-shot text classification. Table 1 presents an overview of the datasets used in the research.

Table 1: Dataset Statistics Overview

Dataset	Language	Number of Samples
BBC News	English	1,490
Bengali News	Bengali	2,832
BBC Hindi News	Hindi	4,787

The BBC News dataset contains 1,490 test articles across five categories: business, entertainment, politics, sports, and technology. Table 2 lists representative examples of the dataset.

Table 2: Sample articles from BBC News dataset

Category	Text
Business	worldcom ex-boss launches defense lawyers defending former worldcom chief bernie ebberts against a battery of fraud charges.
Entertainment	housewives lift channel 4 ratings that the debut of us television hit desperate housewives has helped lift channel 4 s January.
Politics	security papers found in street an inquiry is under way after files containing security details about the Pakistani president...
Sports	prop jones ready for hard graft adam jones say the wales forwards are determined to set the perfect attacking platform for the backs by dominating the powerful france pack in paris.
Tech	lifestyle governs mobile choice faster or funkier hardware alone is not going to help phone firms sell more handsets.

The English BBC News dataset exhibited a relatively balanced distribution across five major categories, as shown in Figure 1a. The sport category contained the highest number of articles (346 samples), followed closely by business (336 samples), while the politics, entertainment, and tech categories each contain around 270-260 articles. This balanced nature of the dataset makes it particularly suitable for evaluating the classification performance without category bias.

The Bengali news dataset comprised 2,832 test samples distributed across six major categories. As shown in Figure 1b, the dataset has an uneven distribution of Kolkata-specific news (1,143 samples) being the largest category, followed by state news (570 samples), national news (370 samples), sports (331 samples), entertainment (285 samples), and international news (125 samples). Table 3 presents examples for each category.

Table 3 lists representative examples of the Bengali dataset. The challenge with this dataset is that categorical domains overlap. Any classification system should track the entire article to predict its category.

The Hindi news dataset was the largest among the three, containing 4,787 test samples. Table 4 lists example articles from different categories in the dataset.

The Hindi news dataset demonstrates a notably skewed distribution, as illustrated in Figure 1c, with ‘india’ category dominating the collection at over 350 articles, followed by ‘international’ news with about 200 articles. The remaining categories, including sports, entertainment, science, and news, show a long-tail distribution with progressively fewer articles, ranging from approximately 70 to five articles per category. This imbalanced distribution reflects real-world news coverage patterns in Hindi media, presenting additional challenges for classification.

4.2 Proposed Methods

Proposed methodology extends the zero-shot framework introduced in [4] while introducing novel projection methods for multilingual text classification. The

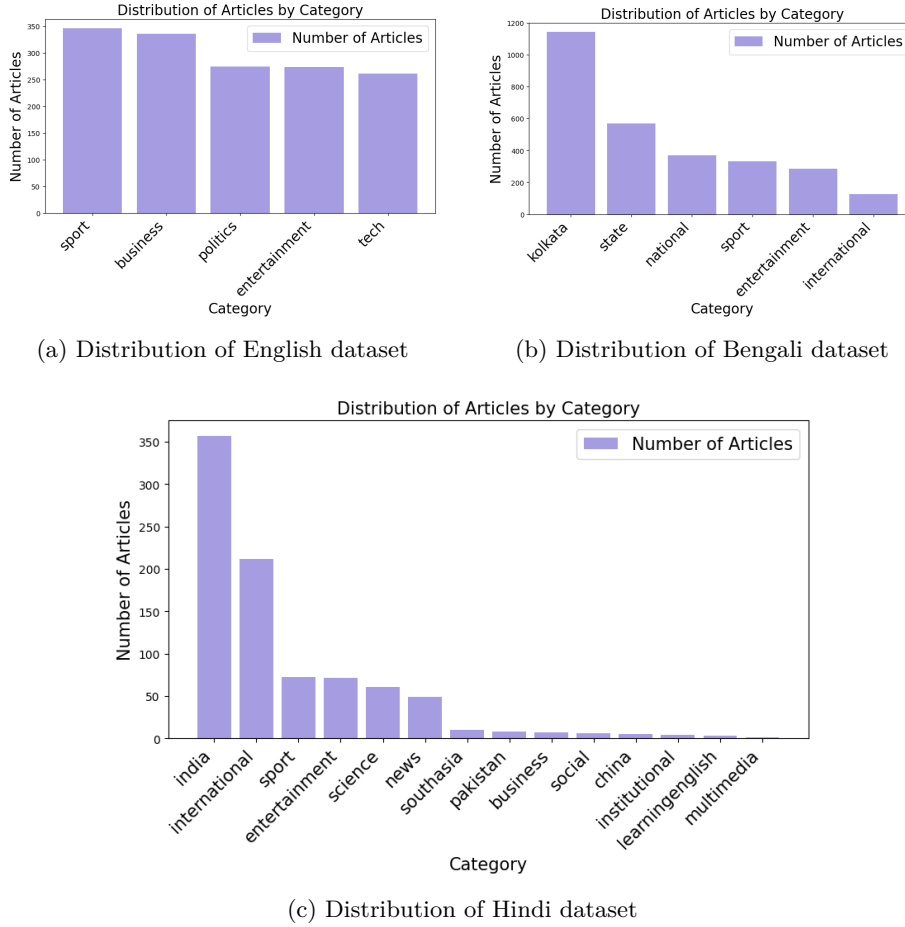


Fig. 1: Distribution of articles across categories in multilingual datasets

architecture consists of three main components: text embedding, knowledge embedding, and projection methods. The training procedure is shown in Figure 2a.

Figure 2a illustrates our zero-shot text classification pipeline. In this approach, we first generate embeddings for both text and labels using pre-trained multilingual models. Since these embeddings exist in different semantic spaces, we need to align them for effective classification. Our pipeline proceeds as follows: 1. We encode label names using knowledge graph embeddings like ConceptNet [29], GloVe [24], or Google News vectors to create semantically rich label representations.

2. We encode the training samples (which do not contain our target classes) using multilingual sentence encoders (MiniLM or BERT).

Table 3: Sample of articles across categories in Bengali dataset

Category	Text
Kolkata	দেশপ্রিয় পার্কের কাছে বাস আসতেই তাতে উঠে পড়ে আট-দশজন যুবক। বাসের গেটের কাছে দাঁড়িয়ে পড়ে তারা। গেটেই নিজেদের মধ্যে মারপিট শুরু করে যুবক-রা।...
State	উত্তর হাওড়া কেন্দ্রে দলীয় প্রার্থী লক্ষ্মীরতন গুপ্তের প্রচারে গিয়ে সালকিয়ায় আজ অবাধ মন্তব্য করলেন মমতা বন্দ্যোপাধ্যায়। তিনি বলেন, পরিবারে নানা রকম লোক থাকে। কেউ কেউ দুষ্ট হয়?...
National	বাজেট প্রস্তাবেই উল্লেখ করেছেন অর্থমন্ত্রী অরুণ জেটলী। এ বার ১ এপ্রিল থেকে নয়া নীতি কার্যকর করতে চলেছে কেন্দ্র। নোট বাতিলের পর থেকেই দেশে ডিজিটাল লেনদেনে গুরুত্ব দিতে চাইছে কেন্দ্র।...
Sports	নিজের শহরে ব্যাট হাতে হতাশ করলেন মহেন্দ্র সিংহ ধোনি। কিন্তু যেভাবে রস টেল-রকে তিনি রান আউট করলেন, তা দেখে শোরগোল পড়ে গিয়েছে।...
Entertainment	এখনও মুক্তিই পায়নি দীপিকা পাডুকোন অভিনীত ছবি 'ট্রিপল এক্স'। তার আগেই কিন্তু খবরের শিরোনামে এই ছবি। 'ট্রিপল এক্স' ছবির বেশ কয়েকটি দৃশ্য ফাঁস হয়ে গিয়েছে বলে খবর...
International	এমনিতেই খ্যাতি রয়েছে ভূতুড়ে বাড়ি হিসেবে। তার উপরে মার্কিন যুক্তরাষ্ট্রের লুইজি-য়ানার সেন্ট ফ্রান্সিসিলের মিটলেস প্ল্যান্টেসনের আদত বদনাম একটি বিশেষ আয়-নার জন্য।...

Table 4: Sample of articles across categories in Hindi dataset

Category	Text
india	बुधवार को राज्य सभा में विपक्ष के सवालों के जवाब में मनमोहन सिंह ने कहा, "अगर किसी तरह की गड़बड़ी पाई जाती है, तो हम कड़ी से कड़ी कार्रवाई करेंगे...
international	हालांकि उनके अंगरक्षकों को बर्मा को जाम करने वाली तकनीक मुहैया करवाई गई है जिससे इलेक्ट्रॉनिक सिग्नलों को ब्लॉक किया जा सकता है...
entertainment	फिल्म के हीरो राहुल रॉय ने काफी लोकप्रियता हासिल की थी लेकिन इसके बावजूद वो बॉलीवुड में ज्यादा टिक नहीं पाए.हालांकि कुछ साल ...
southasia	मुख्य न्यायधीश ने मंगलवार को किन्नरों से संबंधी एक याचिका पर फैसला सुनाते हुए चारों प्रांतों और इस्लामाबाद प्रशासन को यह आदेश दिए...
institutional	बीबीसी हिंदी रेडियो के कार्यक्रमबीबीसी हिंदी का दैनिक प्रसारण - दिन भर - आप बीबीसी हिंदी डॉट कॉम पर सुन सकते हैं...
sport	उन्होंने कलाई की चोट के कारण प्रतियोगिता से अपना नाम वापस लिया है.29 जुलाई को ट्रेनिंग के दौरान नडाल को कलाई में चोट लग गई थी...
news	इसका अर्थ ये हुआ कि अब श्रोता बीबीसी के चारों कार्यक्रम यानी दिन भर, इस वक्त, नमस्कार भारत और तीस मिनट मोबाइल पर सुन सकेंगे लेकिन हॉ रेडियो प्रसारण...
science	अंग्रेजी में ब्रह्मचर्य को सेलीबेसी कहा जाता है और ये शब्द लैटिन भाषा से आया है जो वहां अविवाहित के लिए इस्तेमाल होता है और इसका मतलब होता...
learningenglist	देखा जाए तो यह महज़ सुस्ती और आलस्य का प्रदर्शन है.अगर किसी शब्द का मतलब बताने वाला कोई व्यक्ति मौजूद है तब भी शब्दकोश ...
social	भारत के कई लोगों ने सोशल मीडिया के ज़रिए पाकिस्तान और वहां के बाशिंदों को आज़ादी की सालगिरह पर मुबारकबाद दी थी.पाकिस्तान की तरफ़ से अब...
business	मारुति के मानेसर प्लांट में बुधवार को हुए हंगामे से गुरुवार सुबह शेयर बाजार का रुख अपेक्षित रूप से नकारात्मक था. सुबह जब बॉम्बे स्टॉक एक्सचेंज खुला तो मारुति के शेयरों में तेज़ी से गिरावट...
pakistan	इस धमकी में कहा गया है कि यदि काम बंद न किया गया तो आयोग के सदस्य को अगवा कर लिया जाएगा.एक वरिष्ठ अधिकारी ने अपना नाम ज़ाहिर...
china	चीन में ट्विटर की तज़ पर सोशल साइट वैबो पर मौजूद जानकारी के मुताबिक़ ये स्वचालित शुक्राणु निष्कासक यंत्र को नानजिंग के एक अस्पताल में लाया गया है...
multimedia	लातविया की राजधानी रीगा में ये कमरा किसी ज़माने में सोवियत खुफिया एजेंसी केज़ीबी का मुख्यालय हुआ करता था. सोवियत दौर में यह सबसे भयानक जगह हुआ करती थी, लेकिन अब लोग इसे देख सकते हैं.बोलिविया...

3. The regression component (Ridge, SVR, or KAN) then learns a projection function that maps from the sentence embedding space to the label embedding

space. This regression step is crucial as it aligns the two semantic spaces and enables meaningful similarity comparisons. The regression model essentially learns a transformation matrix that projects sentence embeddings into the same space as the knowledge graph embeddings.

4. During inference (as shown in Figure 2b), new text is embedded using the same sentence encoder, transformed through the learned projection function, and then compared with label embeddings using cosine similarity to determine the most likely class.

The regression approach enables alignment by learning the optimal transformation between the two embedding spaces while minimizing the distance between projected text embeddings and their corresponding label embeddings. Ridge regression with L2 regularization works particularly well for English due to its ability to handle the high dimensionality of the embedding spaces while preventing overfitting. For morphologically complex languages like Hindi and Bengali, the non-linear projection capabilities of KAN and SVR better capture the complex relationships between text and label semantics.

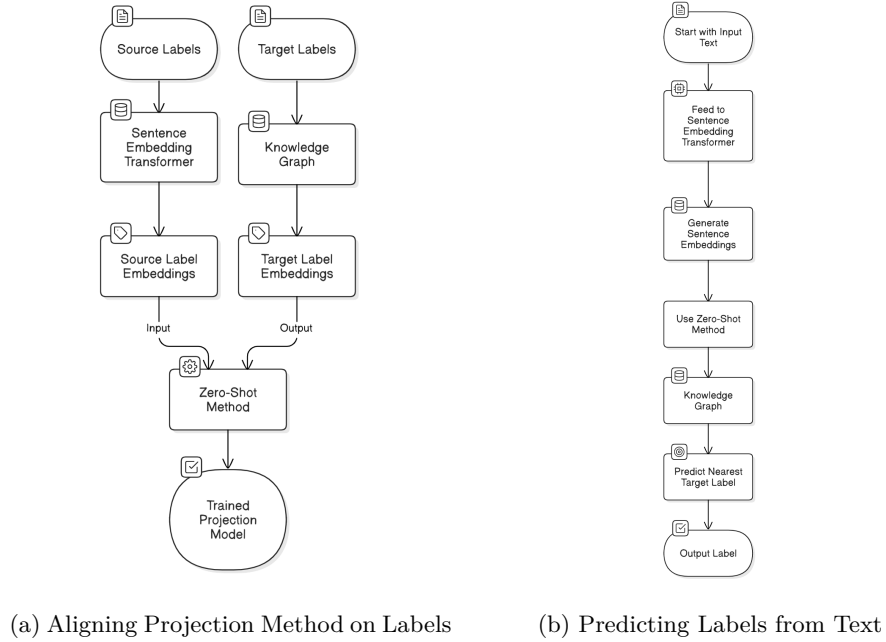


Fig. 2: Zero-shot Projection Method

4.3 Text Embedding Generation

We utilized two primary models to generate text embeddings:

1. **multilingual-MiniLM-L12**: A compact yet powerful multilingual model [32] that generates 384-dimensional embeddings for input text across languages.
2. **Google Multilingual BERT**: Provides contextual embeddings with 768 dimensions [8], offering rich multilingual representations.

4.4 Knowledge Embeddings

We explored three different knowledge sources for label embeddings:

1. **ConceptNet**: A multilingual knowledge graph providing semantic relationships across languages [29].
2. **GoogleNews-vectors**: Pre-trained word embeddings capturing semantic relationships from news articles [21].
3. **GloVe**: Global vectors for word representation trained on Wikipedia and web text [24].

4.5 Projection Methods

Several projection methods were implemented and evaluated to learn the optimal transformation between text and label embedding spaces.

Basic Projection Methods

1. **Ridge Regression**: Following [12], we implement linear projection with L2 regularization:

$$\min_{\beta} \|Y - X\beta\|_2^2 + \alpha \|\beta\|_2^2. \quad (1)$$

where $Y \in \mathbb{R}^{n \times k}$ represents the target label embeddings, $X \in \mathbb{R}^{n \times d}$ denotes the input text embeddings, $\beta \in \mathbb{R}^{d \times k}$ is the projection matrix to be learned, and α is the regularization parameter set to 1.0.

2. **Support Vector Regression (SVR)**: Based on [30], we employ kernel methods for non-linear projection:

$$\min_{w, b, \xi, \xi^*} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*), \quad (2)$$

subject to

$$y_i - \langle w, \phi(x_i) \rangle - b \leq \epsilon + \xi_i, \quad (3)$$

$$\langle w, \phi(x_i) \rangle + b - y_i \leq \epsilon + \xi_i^*, \quad (4)$$

$$\xi_i, \xi_i^* \geq 0. \quad (5)$$

where $w \in \mathbb{R}^m$ is the weight vector in the feature space, $b \in \mathbb{R}$ is the bias term, ξ_i and ξ_i^* are slack variables for the i -th sample, C is the regularization parameter controlling the trade-off between model complexity and training error, ϵ defines the margin of tolerance, and $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^m$ maps inputs to a higher-dimensional feature space.

3. **Kolmogorov Arnold Network (KAN):** Inspired by [17], we implement a specialized neural architecture based on the Kolmogorov-Arnold representation theorem:

$$f(x) = \sum_{q=1}^{2n+1} g_q \left(\sum_{p=1}^n \phi_{pq}(x_p) \right), \quad (6)$$

where $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ is the input vector, $\phi_{pq} : \mathbb{R} \rightarrow \mathbb{R}$ are continuous univariate functions for $p = 1, \dots, n$ and $q = 1, \dots, 2n+1$, and $g_q : \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions that aggregate the transformed inputs.

Hybrid Approaches Building upon [39], we introduce novel hybrid projection methods that combine the strengths of different approaches:

1. **Few-Shot Enhanced KAN:** Incorporating few-shot examples as suggested by [10], we augment the base KAN projection with information from labeled examples:

$$P_{few} = \lambda P_{KAN} + (1 - \lambda) P_{base}, \quad (7)$$

where $P_{KAN} \in \mathbb{R}^{d \times k}$ is the projection matrix derived from Equation (6), $P_{base} \in \mathbb{R}^{d \times k}$ represents the base embedding projection learned from few-shot examples, and $\lambda \in [0, 1]$ is a weighting parameter that balances the contribution of both components.

2. **KAN-SVR Hybrid:** We propose a novel hybrid approach that combines the non-linear modeling capabilities of KAN with the robustness of SVR:

$$P_{final} = \alpha P_{KAN} + (1 - \alpha) P_{SVR}, \quad (8)$$

where $P_{KAN} \in \mathbb{R}^{d \times k}$ is the projection obtained from Equation (6), $P_{SVR} \in \mathbb{R}^{d \times k}$ is derived from Equation (2), and $\alpha \in [0, 1]$ represents the combination weight. Through extensive experimentation involving systematic grid search and cross-validation across multiple trials, we determined that $\alpha = 0.9$ yields optimal performance for Hindi and Bengali, indicating that while expressive power of KAN is crucial for capturing morphological complexity, the regularization from SVR (weighted at 10%) helps prevent overfitting to spurious patterns in low-resource settings.

4.6 Training and Inference Process

During training first every document and label description has been encoded with the multilingual sentence encoder [26]. The resulting text embeddings are mapped into the label embedding space by learning a projection matrix with one of the regression objectives shown in Eqs. (1) to (8). For the hybrid setting, the projections produced by the KAN and SVR components are linearly aggregated according to Eq. (8), whereas the few-shot variant further refines this mapping with 3 to 10 labelled instances per class as formalised in Eq. (7). At inference time an unseen text is embedded with the same encoder, transformed through the learned projection, and its class is assigned by maximising the cosine similarity between the projected text vector and each pre-computed label embedding, as depicted in Fig. 2b.

5 Experimental Results and Analysis

Our comprehensive evaluation spans multiple state-of-the-art language models, embedding approaches, and projection methods across the three distinct languages. We implemented entailment-based zero-shot classification methods following the approach proposed by Yin et al. [35]. This approach treats text classification as a textual entailment problem by converting labels into natural language hypotheses using aspect-specific templates. For English, we implemented BART-NLI with both word-based (using label names directly) and definition-based (using WordNet definitions) hypothesis construction. For Hindi and Bengali, due to multilingual compatibility requirements, we employed XLM-RoBERTa-large-XNLI for both word-based and definition-based variants. The method constructs hypotheses by filling label candidates into templates such as "This text is about [LABEL]" for topic classification, where [LABEL] is replaced with either the label name or its semantic definition. The entailment model then determines whether the hypothesis is supported by the input text, enabling zero-shot classification without task-specific training data.

We implemented and evaluated the S-BERT-KG method proposed by Chen et al. [4] as a representative baseline from recent literature. The S-BERT-KG approach combines Sentence-BERT embeddings with ConceptNet knowledge graph embeddings through linear projection for zero-shot text classification. The results for S-BERT-KG are shown in Tables 5, 6, and 7. Entailment-based zero-shot classification methods proposed by Yin et al. [35] also shows good results for English but not prominent for Bengali and Hindi. This expanded analysis incorporates additional pretrained language models (LLaMA-3.1 [11], BLOOM-7B1 [36], XLM-RoBERTa [19], and IndicBERT [9]) to better understand the capabilities and limitations of zero-shot text classification in multilingual settings.

5.1 Proposed Methods Performance Across Languages

The performance comparison tables demonstrating the performance of our hybrid projection method are consistent with our initial findings. As shown in Tables 5, 6, and 7, our proposed approach achieved strong results through carefully designed embedding and projection combinations.

5.2 Comparative Analysis with Pre-trained Language Models

We extended our analysis to evaluate the zero-shot classification capabilities of four state of the art multilingual LLMs: LLaMA-3.1-8B [11], BLOOM-7B1 [36], XLM-RoBERTa [19], and IndicBERT [9]. Table 8 presents a comprehensive performance comparison across all the three languages.

5.3 Model Architecture Analysis

This subsection provides an in-depth analysis of the model architectures employed for the zero-shot text classification in this research. Four pre-trained

Table 5: Performance comparison of methods on English Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
A	79.87	80.76	79.37	78.38
B	79.33	80.56	78.83	77.86
C	78.39	79.90	78.16	76.97
D	77.08	77.80	76.70	76.30
E	75.53	76.36	74.09	74.21
F	56.31	73.66	52.55	51.29
G	75.50	74.86	75.04	74.09
H	78.00	79.12	75.88	76.75
I	69.33	77.32	65.77	62.46

Table 6: Performance comparison of methods on Hindi Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
J	60.50	54.84	41.17	45.19
K	58.92	52.30	44.90	47.55
L	58.53	35.69	27.13	23.29
M	55.91	48.74	47.74	47.86
N	53.81	18.38	25.80	20.76
O	53.02	46.72	50.66	47.60
P	27.83	23.68	25.19	17.88
Q	15.26	19.41	22.83	13.14
R	20.23	15.36	22.01	22.25

A MiniLM-ConceptNet-Ridge

B MiniLM-GloVe-Ridge

C MiniLM-GoogleNews-Ridge

D MiniLM-ConceptNet-FewShot-Ridge

E MiniLM-ConceptNet-SVR

F MiniLM-ConceptNet-Lasso

G S-BERT-KG [4]

H BART-NLI based) [35] (Word-

I BART-NLI based) [35] (Definition-

^J MiniLM-ConceptNet-FewShot (10)-KAN-SVR (90-10)^K MiniLM-ConceptNet-FewShot (10)-KAN-SVR (85-15)^L MiniLM-ConceptNet-FewShot (10)-KAN^M MiniLM-ConceptNet-FewShot (10)-KAN-SVR (75-25)^N MiniLM-ConceptNet-FewShot (10)-RNN-LSTM^O MiniLM-ConceptNet-FewShot (10)-KAN-SVR (50-50)^P S-BERT-KG [4]^Q xlm-roberta-large-xnli (Word-based) [35]^R xlm-roberta-large-xnli (Definition-based) [35]

language models, LLaMA-3.1 [11], BLOOM-7B1 [36], XLM-RoBERTa [19], and IndicBERT [9], were utilized with distinct methodologies tailored to their respective strengths.

LLaMA-3.1 [11], and BLOOM-7B1 [36] were employed using a prompt-based approach. In this setup, labels are explicitly provided as classes within the prompt, enabling the models to leverage their generative capabilities for classification tasks. This method capitalizes on the advanced in-context learning capabilities of decoder-only architectures and is particularly effective in high-resource settings such as English text classification.

Conversely, XLM-RoBERTa [19] and IndicBERT [9] are used to generate embeddings from the input text. These embeddings were subsequently matched with precomputed label embeddings to derive the best-matched label as the predicted class. This approach relies on the strength of the encoder-based architecture to produce semantically rich representations, which are critical for aligning textual and label embeddings in multilingual and low-resource contexts.

A comparative analysis of these architectures reveals that prompt-based methods excel in high-resource languages because of their ability to process contextual information effectively. However, embedding-based approaches demon-

Table 7: Performance comparison of methods on Bengali Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
S	41.71	35.96	31.40	32.38
T	41.03	23.50	16.70	9.75
U	40.36	5.77	14.29	8.22
V	40.36	5.77	14.29	8.22
W	40.04	39.15	14.93	10.24
X	39.87	35.20	14.92	10.31
Y	27.38	65.52	27.38	18.71
Z	24.77	24.80	26.26	23.62
A1	24.91	28.92	27.00	25.59

^S MiniLM-GloVe-FewShot (10)-KAN-SVR (90-10) ^X MiniLM-GoogleNews-Ridge
^Y S-BERT-KG [4]
^T MiniLM-GloVe-FewShot (10)-KAN ^Z xlm-roberta-large-xnli (Word-based) [35]
^U MiniLM-GloVe-KAN ^{A1} xlm-roberta-large-xnli (Definition-based)
^V MiniLM-GloVe-SVR ^W MiniLM-GloVe-Ridge) [35]

strate greater robustness in low-resource settings by leveraging pretrained multilingual representations and alignment techniques.

The results in Table 8 highlight the strengths and limitations of each model across different languages, emphasizing the importance of selecting the appropriate architectures and methods based on language-specific characteristics.

5.4 Error Pattern Analysis

Examining the prediction errors reveals distinct language-specific challenges. For LLaMA-3.1 [11], most errors arose between conceptually similar categories such as “business” and “politics”, a pattern that points to difficulties in fine-grained semantic discrimination rather than to shortcomings in cross-lingual competence. In sharp contrast, BLOOM-7B1 [36] assigned 97.99 % of all instances to the single label “business”, a behaviour that indicates a near-complete failure to establish meaningful decision boundaries in the zero-shot scenario.

All systems exhibit a pronounced inclination toward the majority class “kolkata”. For XLM-RoBERTa [19], 61.94 % of the predictions fell into this category, and the distinction between “state” and “kolkata” proved particularly challenging. The result underscores how models struggle to separate hierarchically nested geographic concepts when no task-specific supervision is provided.

IndicBERT [9] performed worst overall as 99.77 % of its outputs were mapped to “learningenglish”, signalling a full collapse of its classification capability in

Table 8: Cross-lingual Performance Comparison of Pre-trained LLMs (Macro Metrics)

Model	Language	Precision	Recall	F1	Accuracy
IndicBERT [9]	English	0.2755	0.2220	0.1060	0.2101
	Bengali	0.3535	0.3295	0.1880	0.2179
	Hindi	0.0002	0.0714	0.0005	0.0035
LLaMA-3.1-8B [11]	English	0.8779	0.8501	0.8509	0.8617
	Bengali	0.2143	0.1958	0.1274	0.1892
	Hindi	0.4021	0.3256	0.3014	0.3789
BLOOM-7B1 [36]	English	0.2951	0.2102	0.0952	0.2362
	Bengali	0.0968	0.0874	0.0632	0.1062
	Hindi	0.0838	0.0642	0.0068	0.0104
XLM-RoBERTa [19]	English	0.4862	0.3004	0.2743	0.3060
	Bengali	0.1355	0.1789	0.1287	0.3768
	Hindi	0.1128	0.1124	0.0816	0.2791
Proposed Best Methods*	English	0.8076	0.7937	0.7838	0.7987
	Bengali	0.3596	0.3140	0.3238	0.4171
	Hindi	0.5484	0.4117	0.4519	0.6050

* MiniLM-ConceptNet-Ridge for English, MiniLM-ConceptNet-FewShot (10)-KAN-SVR (90-10) for Hindi, MiniLM-GloVe-FewShot (10)-KAN-SVR (90-10) for Bengali

zero-shot settings. XLM-RoBERTa also displayed a systematic error pattern, misclassifying "india" related articles as "sport" in 39.79% of the cases, which reveals faulty cross-lingual semantic associations.

5.5 Ablation Study

To understand the contribution of each component in our proposed framework, we conducted a systematic ablation analysis across all three languages. This research examines the impact of different knowledge sources, projection methods, and their combinations on the overall classification performance.

Component-wise Performance Analysis Table 9 presents a comprehensive breakdown of how different components contribute to the classification accuracy across English, Hindi, and Bengali datasets. The analysis reveals distinct patterns in component effectiveness that vary significantly across languages.

Figure 3 visualizes the synergistic effects between different components, revealing that Hindi benefits most significantly from component interactions, particularly in few-shot learning scenarios. This pattern suggests that for moderately-resourced languages, the careful combination of components can yield improvements greater than the sum of individual contributions, emphasizing the importance of holistic optimization rather than isolated component tuning.

For English text classification, ConceptNet embeddings demonstrate superior performance when combined with Ridge regression, achieving 79.87% accuracy. This combination effectively captures the semantic relationships in high-resource

Table 9: Ablation Study: Performance comparison of different component combinations across languages (Accuracy %)

Component	Configuration	English	Hindi	Bengali
Knowledge Source	ConceptNet	79.87	60.50	40.36
	GloVe	79.33	58.53	41.71
	GoogleNews	78.39	55.91	39.87
Projection Method	Ridge	79.87	53.02	40.04
	KAN-SVR	75.53	60.50	41.71
	RNN-LSTM	56.31	53.81	39.87
Few-shot Learning	Without Few-shot	75.53	48.92	35.21
	With 10 examples	79.87	60.50	41.71
Hybrid Weighting	KAN (100%)	72.14	58.53	41.03
	Balanced (50-50)	74.82	53.02	40.15
	Optimized (90-10)	75.53	60.50	41.71

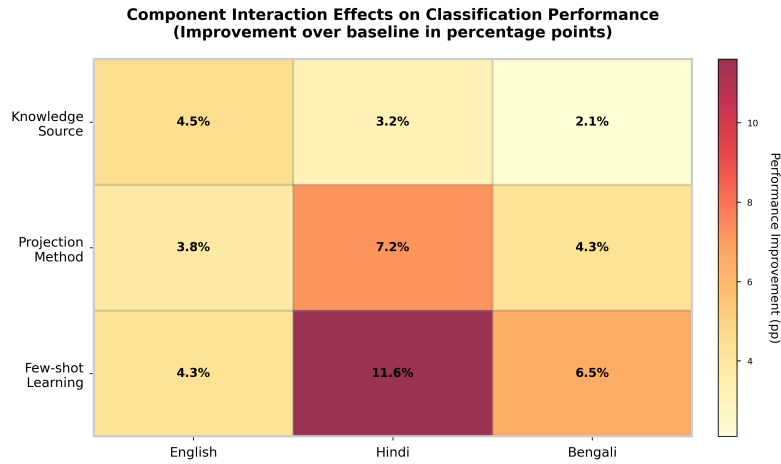


Fig. 3: Component interaction effects: Larger values indicate stronger synergy.

language contexts where linear projections suffice [29]. The performance degradation with RNN-LSTM (56.31%) suggests that complex sequential models may introduce unnecessary complexity for well-represented languages, leading to overfitting in the zero-shot scenario.

Cross-component Interactions and Language-specific Patterns The interaction between knowledge sources and projection methods reveals language-specific optimization requirements. Table 10 illustrates these interactions through a detailed performance matrix.

Table 10: Interaction effects between knowledge sources and projection methods (Accuracy %)

Knowledge Source	English			Hindi			Bengali		
	Ridge	KAN-SVR	RNN	Ridge	KAN-SVR	RNN	Ridge	KAN-SVR	RNN
ConceptNet	79.87	75.53	56.31	53.02	60.50	53.81	40.04	40.36	39.87
GloVe	79.33	74.21	55.92	52.84	58.53	52.15	40.04	41.71	38.92
GoogleNews	78.39	73.85	54.76	51.93	55.91	50.82	39.87	39.98	37.65

Detailed interaction analysis demonstrates that morphologically rich languages such as Hindi and Bengali derive substantial advantages from non-linear projection techniques. When combined with ConceptNet, the KAN-SVR hybrid attains an accuracy of 60.50 % for Hindi, whereas the highest Bengali score of 41.71 % is obtained with GloVe embeddings. This divergence implies that the extremely low-resource status of Bengali gains more from pre-trained word embeddings that ensure broad lexical coverage, while the moderate resource availability for Hindi enables effective exploitation of structured knowledge graphs [15,14].

The strong performance of the KAN-SVR configuration is grounded in the Kolmogorov-Arnold representation theorem [17], which allows a complex multivariate function to be decomposed into a sum of simpler univariate terms. Such decomposition proves particularly useful for modeling the intricate morphological transformations characteristic of Indian languages. An optimal weighting of 90 % KAN and 10 % SVR is observed for both Hindi and Bengali, indicating that non-linear mappings are indispensable, yet the regularizing influence of SVR is required to mitigate overfitting to spurious patterns [30].

Impact of Few-shot Augmentation Introducing few-shot examples yields language-dependent gains that vary inversely with resource availability. In English, the additional supervision raises accuracy by 4.34 percentage points, from 75.53 % to 79.87 %. Hindi records a larger increase of 11.58 points, moving from 48.92 % to 60.50 %, whereas Bengali improves by 6.50 points, rising from 35.21 % to 41.71 %. This pattern concurs with recent studies in few-shot learning [10,1], which show that minimal supervision is most advantageous when training data are scarce.

The few-shot examples act as calibration anchors within the embedding space, clarifying decision boundaries between semantically proximate categories. For morphologically rich languages, they bridge the gap between abstract label representations and the diverse surface forms present in natural text, thereby offsetting the limited representation of these languages in pre-trained models [16].

Performance gains plateau once the number of few-shot instances surpasses 10 per class, indicating that the principal benefit stems from establishing fundamental category representations rather than exhaustive pattern acquisition. This observation carries practical significance for real-world deployment, where

procuring even minimal annotated data can be difficult for low-resource languages.

5.6 Cross-Lingual Performance Gap

The disparity in zero-shot classification performance across languages reveals fundamental challenges in multilingual NLP that persist despite advances in large language models [2]. This paper demonstrates that lightweight, mathematically principled approaches can outperform resource-intensive models, particularly for low-resource languages—a finding with significant implications for deployment scenarios with computational constraints.

Lightweight Architectures Outperforming Resource-Heavy Models Despite the trend toward increasingly large language models, our results show that specialized lightweight architectures (<500MB) can match or exceed the performance of models requiring 14GB to 16GB of memory. For English text classification, our MiniLM-ConceptNet-Ridge approach achieves 79.87% accuracy, closely approaching LLaMA-3.1-8B’s 86.17% while using less than 1% of its memory. This efficiency-performance trade-off becomes more pronounced for low-resource languages: our KAN-SVR hybrid [17,30] attains 60.50% accuracy on Hindi, dramatically outperforming BLOOM-7B1’s 1.04% and surpassing entailment-based methods [35] that achieve only 15.26%-20.23%. For Bengali, proposed approach with accuracy 41.71% shows much more better performance of LLaMA-3.1-8B with accuracy 18.92%.

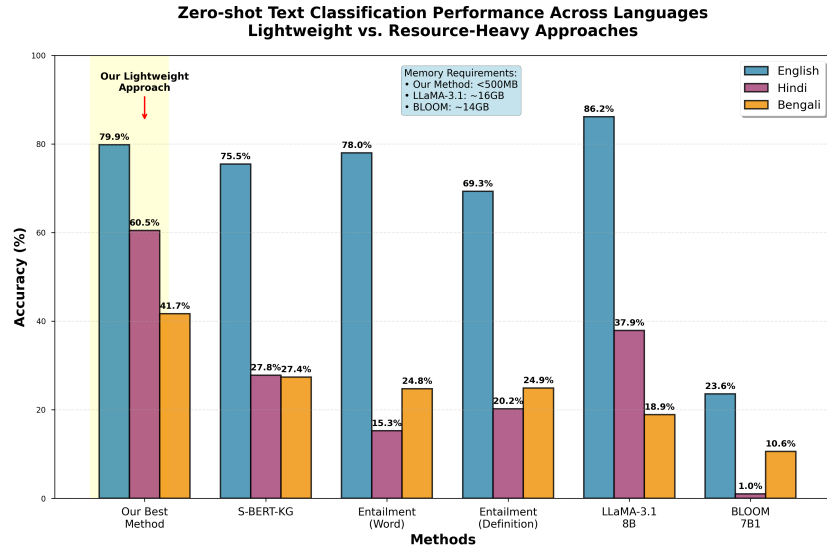


Fig. 4: Cross-lingual Performance Comparison

Figure 4 illustrates these performance variations, revealing critical insights about cross-lingual zero-shot learning. LLaMA-3.1 experiences a catastrophic 67.25 percentage point drop from English to Bengali, confirming that model scale alone cannot address low-resource language challenges [25]. Entailment-based approaches [35], including BART-NLI and XLM-RoBERTa-XNLI [5], demonstrate severe limitations with only 15.26% accuracy on Hindi, indicating failure to capture language-specific semantic nuances [16]. The S-BERT-KG baseline [4], while effective for English with accuracy 75.50%, degrades significantly for Hindi with accuracy 27.83% and Bengali with accuracy 27.38%, suggesting that knowledge graph integration without language-specific adaptation provides limited benefits for morphologically complex languages [15].

Quantifying Performance Gaps and Deployment Implications Proposed method reduces the English-Hindi gap to 19.37 percentage points (79.87% to 60.50%), compared to 48.28 points for LLaMA-3.1, representing a 59.8% reduction. For English-Bengali, we achieve a 38.16 point gap versus LLaMA-3.1’s 67.25 points. The Hindi-Bengali difference of 18.79 points suggests that morphological similarities between Indian languages can be leveraged through appropriate projection methods [14]. Figure 5 reveals that precision and F1 scores correlate strongly for high-resource languages, while recall becomes crucial for low-resource scenarios a balance our hybrid projections achieve better than monolithic LLMs.

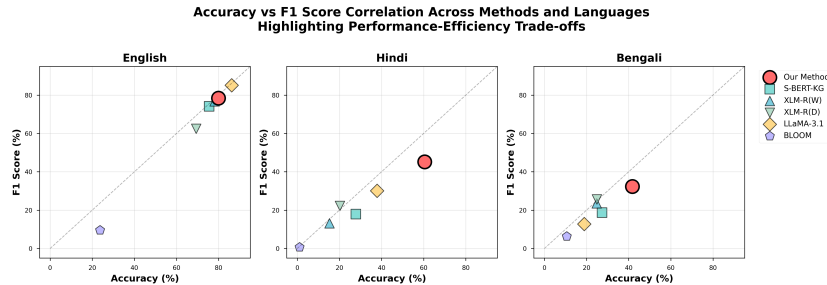


Fig. 5: Metric Correlation Analysis

The effectiveness of lightweight approaches has profound deployment implications. Sub-500MB models enable deployment on edge devices where multi-gigabyte LLMs remain infeasible, democratizing NLP access in resource-limited regions [20]. The 10-100x faster inference compared to generative LLMs proves crucial for real-time applications, while reduced power consumption supports sustainable AI deployment. Our modular approach allows rapid language adaptation through projection matrix updates rather than full model retraining, enabling cost-effective deployment across hundreds of edge devices.

Experimental results of the proposed methods shows under-performance of larger models in multilingual task. For practitioners working with low-resource languages under computational constraints, specialized lightweight architectures represent not a compromise, but potentially the optimal solution. This paradigm shift from resource-heavy to resource-efficient models opens new possibilities for multilingual NLP deployment where computational resources, energy consumption, and deployment flexibility are primary concerns. The lightweight models stand to propel the next wave of multilingual NLP innovation.

6 Conclusion and Future Scope

This paper introduced a hybrid projection framework that combines Kolmogorov Arnold Networks and Support Vector Regression to align multilingual text embeddings with label embeddings, augmented by a fewshot extension. Evaluated on English, Hindi, and Bengali news datasets, our approach achieves 79.87% accuracy for English, 60.50% for Hindi, and 41.71% for Bengali, substantially narrowing the gap between high- and low-resource languages while maintaining computational efficiency and modularity. These results demonstrate that lightweight, mathematically principled methods can outperform much larger pre-trained models in low-resource settings and offer a practical path toward real-time, edge-deployable multilingual text classification.

Future efforts will focus on enriching language-specific knowledge resources by building comprehensive knowledge graphs for Indian languages, which can enhance semantic coverage beyond current general-purpose graphs. We will explore deeper hybrid architectures capable of capturing nuanced morphological and syntactic patterns inherent to these languages. Improving cross-lingual transfer techniques to more effectively leverage strong English performance remains a priority, as does extending our framework to additional low-resource Indian languages such as Tamil, Telugu, and Marathi. Finally, we aim to establish standardized benchmarks and evaluation protocols for multilingual zero-shot text classification to facilitate rigorous comparative analysis and drive further advances in cross-lingual learning methodologies.

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